Chapter 19 Review

Campbell Biology, AP Edition, 12e

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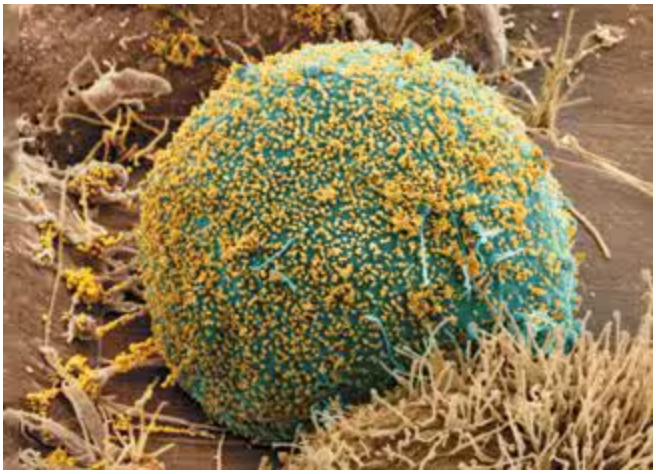


Summary of Key Concepts

Does the horizontal transfer of genes by viruses create problems for scientists who use gene sequences to determine evolutionary relationships? Justify your answer. **(BIG IDEAS 1 & 3)**

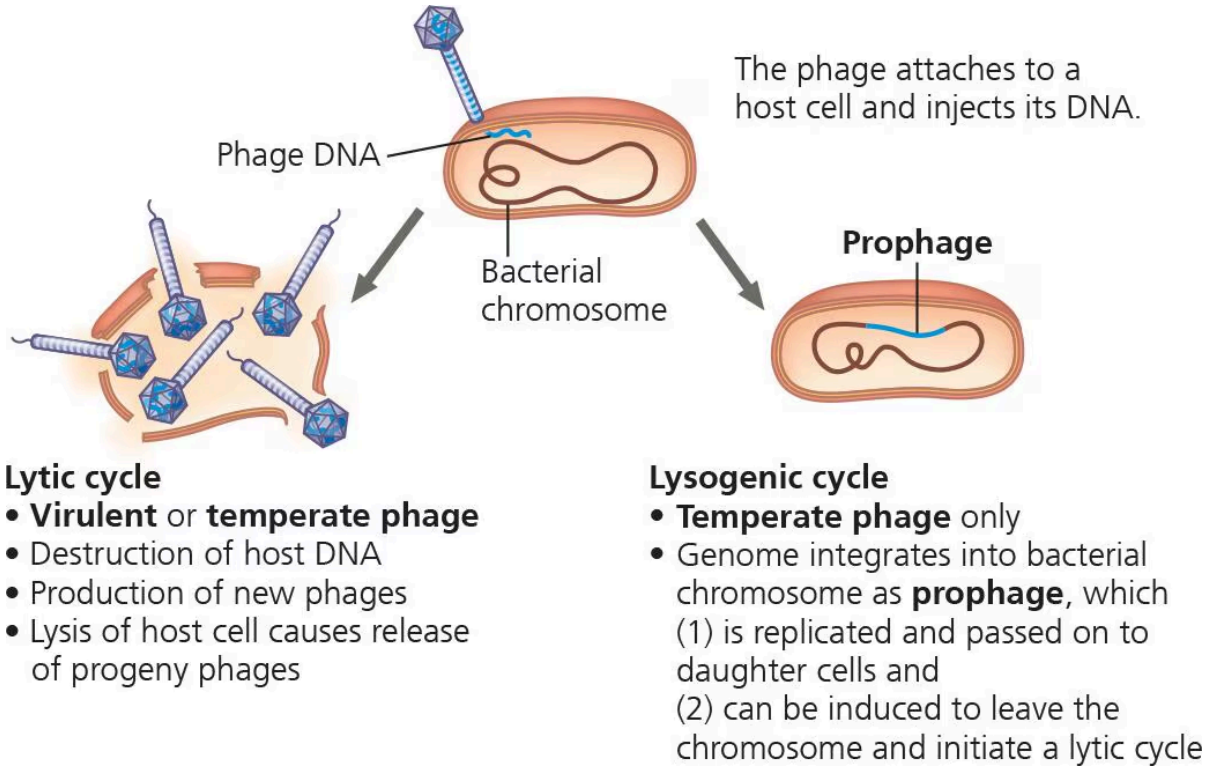
Concept 19.1: A virus consists of a nucleic acid surrounded by a protein coat

- Researchers discovered viruses in the late 1800s by studying a plant disease, tobacco mosaic disease.
- A **virus** is a small nucleic acid genome enclosed in a protein **capsid** and sometimes a membranous **viral envelope**. The genome may be single- or double-stranded DNA or RNA.

Are viruses generally considered living or nonliving? Explain.

Concept 19.2: Viruses replicate only in host cells

- Viruses use enzymes, ribosomes, and small molecules of host cells to synthesize progeny viruses during replication.
- Each type of virus has a characteristic ****host range,** affected by whether cell-surface proteins are present that viral surface proteins can bind to.
- **Phages** (viruses that infect bacteria) can replicate by two alternative mechanisms: the **lytic cycle** and the **lysogenic cycle**.



The phage attaches to a host cell and injects its D N A. This induces either the lytic or lysogenic cycle.

Lytic cycle:

- Virulent or temperate phage

- Destruction of host D N A
- Production of new phages
- Lysis of host cell causes release of progeny phages

Lysogenic cycle:

- Temperate phage only
- Genome integrates into bacterial chromosome as prophage, which (1) is replicated and passed on to daughter cells and (2) can be induced to leave the chromosome and initiate a lytic cycle.
- Bacteria have various ways of defending themselves against phage infections, including the CRISPR-Cas system.
- Many animal viruses have an envelope. **Retroviruses** (such as **HIV**) use the enzyme ****reverse transcriptase** to copy their RNA genome into DNA, which can be integrated into the host genome as a **provirus**.
- Since viruses can replicate only within cells, they probably evolved after the first cells appeared, perhaps as packaged fragments of cellular nucleic acid.

Describe enzymes that are not found in most cells but are necessary for the replication of certain types of viruses.

Concept 19.3: Viruses and prions are formidable pathogens in animals and plants

- Symptoms of viral diseases may be caused by direct viral harm to cells or by the body's immune response. ****Vaccines** stimulate the immune system to defend the host against specific viruses.
- An **epidemic**, a widespread outbreak of a disease, can become a ****pandemic**, a global epidemic.
- Outbreaks of emerging viral diseases in humans are usually not new, but rather are caused by existing viruses that expand their host territory. The H1N1 2009 flu virus was a new combination of pig, human, and avian viral genes that caused a pandemic. The H5N1 avian flu virus has the potential to cause a high-mortality flu pandemic.
- Viruses enter plant cells through damaged cell walls (horizontal transmission) or are inherited from a parent (vertical transmission).
- ****Prions** are slow-acting, virtually indestructible infectious proteins that cause brain diseases in mammals.

What aspect of an RNA virus makes it more likely than a DNA virus to become an emerging virus?